

ARESH POURKAVOOS

apour3@uic.edu · mathtician.github.io

EDUCATION

U. of Illinois Chicago (UIC)

Chicago, IL

MS, Computer Science

Cumulative GPA: 4.00

Aug 2025 - May 2027

Carnegie Mellon Uni. (CMU)

Pittsburgh, PA

BS, Computer Science

Minor, Math

Cumulative GPA: 3.83

Aug 2021 - May 2025

Loomis Chaffee (LC)

Windsor, CT

Sep 2017 - Jun 2021

SELECTED COURSES

UIC

CS 461 Operating Systems

CS 494 Special Topic: Agentic AI

CS 594 Special Topic: RLHF

CS 463 Systems Performance

CMU

10-701 Intro to ML (PhD)

- Group project: review of language model compression with experiments

15-411 Intro to Compiler Design

- Pair programmed a compiler in Rust for a subset of C
- Wrote an x86 assembler for final project

15-312 Foundations of PL

21-238 Honors Abstract Alg. II

21-237 Honors Abstract Alg. I

21-235 Honors Real Analysis I

SKILLS

Programming Languages

Python, Rust, OCaml, C, C++

Technologies

Linux, Git, PyTorch, Codex,

Claude Code, llama.cpp,

OpenCV

WORK EXPERIENCE

Computer Vision Engineer

Alphacore, Inc.

Summer 2025

- Investigated Git self-hosting for internal software
- Designed a stereo vision pipeline for long-range maritime multi-object tracking
- Built prototype with GUI for parameter tuning
 - OpenCV, YOLOv11, Dear ImGui, SLAM, Kalman filters

PROJECTS

Autoformalization

Lean, Claude Code, MCP

- Formalize a major result from the paper "Weak Memory Models with Matching Axiomatic and Operational Definitions" (Zhang et al, 2017)
- Repo: github.com/mathtician/gam-formalization

Idea book

LaTeX

- Maintaining repository of math ideas
- Repo: github.com/mathtician/ideas

RESEARCH

Formal Methods Research

UIC

Summer 2026

- Began thesis research with Prof. William Mansky on concurrent verified compilers

Machine Learning Research

CMU

Summer 2024

- Worked with Prof. Andrej Risteski towards bounding the convergence rate of denoising score matching

ACTIVITIES

15-317 Course Staff

CMU

Teaching Assistant

Aug-Dec 2024

- Recitations, office hours, grading

CMU Computer Club

CMU

President

Jan-Dec 2024

- Organize and advertise club events
- Help maintain club-hosted servers, used by many student orgs

AWARDS

Top 200

William Lowell Putnam Competition, Dec 2022

Top 500

Putnam Competition, Dec 2021

MISC

US Citizen

